E-commerce Advanced SQL Project

Project Overview

This project focuses on analyzing an e-commerce dataset using SQL. It simulates a real-world business environment by designing a relational database, loading structured data, and performing advanced analysis to generate meaningful business insights.

The project covers the complete workflow from database creation to solving business problems such as sales analysis, customer behavior, product performance, and operational efficiency.

Business Problem

The e-commerce business generates large volumes of transactional data across multiple tables such as customers, orders, products, and payments. However, raw data alone does not provide actionable insights.

The key challenges addressed in this project include:

- Identifying top-performing products and categories driving revenue
- Understanding customer purchasing behavior and lifetime value
- Detecting high return rates affecting profitability
- Evaluating seller performance and identifying inactive sellers
- Monitoring inventory levels to avoid stock shortages
- Analyzing payment and delivery success rates for operational efficiency

Objectives

- Design a normalized relational database for an e-commerce system
- Load and manage structured data using SQL
- Perform advanced SQL queries to analyze data
- Solve real-world business problems
- Generate insights for data-driven decision-making

Database Design

The database consists of 9 interconnected tables:

- categories – shape(5,2)
- customers – shape(3000,4)
- orders – shape(20000,5)
- order_items – shape(20000,6)
- products – shape(250,5)
- payments – shape(20000,5)
- shipping – shape(20000,5)
- sellers – shape(120,2)
- inventory – shape(250,5)

Each table is connected using Primary Keys and Foreign Keys to maintain data integrity and relationships between entities

SQL Techniques Used

- Joins (INNER, LEFT, RIGHT)
- Aggregations (SUM, COUNT, AVG)
- Common Table Expressions (CTEs)
- Window Functions (RANK, DENSE_RANK, ROW_NUMBER, LAG)
- Subqueries
- CASE statements
- Date functions

These techniques were used to perform complex data analysis and extract insights.

Business Analysis

Product Performance Analysis

This section focuses on identifying top-performing products based on sales and order volume. It helps in understanding which products contribute the most to overall revenue.

1. Top 10 selling products

	product_name character varying (30)	count bigint	total_sales numeric
1	Treadmill	106	990702
2	Kurta	91	912804
3	Face Cream	94	901561
4	Gloves	128	900652
5	Hair Dryer	87	866160
6	Mirror	81	850608
7	Dress	84	841620
8	Broom	97	831422
9	Smart Doorbell	86	823826
10	Extension Board	84	820704

2. Product ranking using window function

	product_name character varying (30)	total_orders bigint	total_sales numeric	rank bigint
1	Treadmill	106	990702	1
2	Kurta	91	912804	2
3	Face Cream	94	901561	3
4	Gloves	128	900652	4
5	Hair Dryer	87	866160	5
6	Mirror	81	850608	6
7	Dress	84	841620	7
8	Broom	97	831422	8
9	Smart Doorbell	86	823826	9
10	Extension Board	84	820704	10

Category Analysis

This section focuses on analyzing the performance of different product categories in terms of revenue contribution and return behavior. It helps in identifying which categories are driving business growth and which are facing issues like high return rates.

1. Best selling category by state

	state character varying (20)	category_name character varying (15)	total_sales numeric	rank bigint
1	Arunachal Pradesh	Beauty	855031	1
2	Chhattisgarh	Clothing	1088413	1
3	Kerala	Electronics	854563	1
4	Kerala	Sports	865994	1
5	Odisha	Home	929851	1

Revenue Analysis

This section analyzes revenue distribution across different categories and compares sales performance over time.

1. Revenue by category

	category_name character varying (15)	total_revenue numeric	contribution numeric
1	Sports	19675913	19.26
2	Electronics	19470982	19.05
3	Home	21653018	21.19
4	Clothing	22679841	22.20
5	Beauty	18704284	18.30

2. 6_months -over- 6_months sales comparison(last vs previous)

	category_name character varying (15)	product_name character varying (30)	current_6m_sales numeric	prev_6m_sales numeric	sales_diff numeric
1	Home	Water Bottle	376831	60672	316159
2	Home	Water Bottle	376831	179088	197743
3	Electronics	Hard Disk	485056	312488	172568
4	Home	Utensil Set	399252	228144	171108
5	Electronics	Smart Doorbell	495248	328578	166670
6	Beauty	Lipstick	488320	327000	161320
7	Sports	Camping Tent	460845	302841	158004
8	Home	Mirror	494208	356400	137808
9	Clothing	Leather Jacket	376556	241487	135069
10	Beauty	Blush	317310	194919	122391
11	Home	Bedsheet	305807	190016	115791
12	Home	Carpet	424299	312128	112171

Customer Analysis

This section evaluates customer behavior, spending patterns, and segmentation to identify high-value customers.

1. Average order value

	name character varying (25)	count bigint	avg numeric
1	Naksh Chacko	13	5499.5384615384615385
2	Veda Pathak	7	4194.5714285714285714
3	Ranveer Kumer	6	4382.6666666666666667
4	Anirudh Morar	6	4282.8333333333333333
5	Urvashi Sani	6	3364.1666666666666667
6	Abdul Pant	9	5106.3333333333333333
7	Vaishnavi Bahri	8	3264.7500000000000000
8	Chakradev Bora	10	6939.4000000000000000
9	Bhavna Narayan	6	7274.0000000000000000
10	Manbir Dave	6	4905.0000000000000000
11	Jeet Narasimhan	7	4532.1428571428571429

2. Customer with no order

	cust_id [PK] character varying (10)	name character varying (25)	state character varying (20)
1	CUST_1252	Adweta Chaudhuri	Karnataka
2	CUST_2262	Janaki Venkataraman	Telangana
3	CUST_2731	Suhani Bera	Meghalaya

3. Customer lifetime value (CLV)

	name character varying (25)	total_value numeric	cx_ranking bigint
1	Upkaar Magar	130000	1
2	Radhika Raju	103314	2
3	Ubika Palla	98903	3
4	Pratyush Nair	91052	4
5	Hredhaan Sachdeva	90690	5
6	Vedant Magar	90604	6
7	Yutika Sachar	89151	7
8	Tanay Choudhary	88362	8
9	Jagat Malhotra	88056	9
10	Warhi Rana	86661	10
11	Nandini Savant	85738	11
12	Jagvi Swamy	84141	12

4. Customer segmentation (Returned vs New

	cust_id [PK] character varying (10)	name character varying (25)	total_orders bigint	returned_orders bigint	case text
1	CUST_2400	Nathan Agarwal	6	2	new
2	CUST_2462	Ishanvi Thakur	10	1	new
3	CUST_1216	Victor Dixit	8	4	new
4	CUST_2761	Arya Konda	4	1	new
5	CUST_1360	Ekani Anand	10	3	new
6	CUST_2221	Warhi Mahajan	7	2	new
7	CUST_2776	Kashish Dugal	6	1	new
8	CUST_2114	Libni Shere	4	0	new
9	CUST_1602	Vasana Yadav	6	3	new
10	CUST_2478	Kashish Sule	9	4	new
11	CUST_2902	Anika Master	7	2	new

5. Top customer by state

	state character varying (20)	name character varying (25)	total_orders bigint	total_sales numeric	rn bigint
1	Andhra Pradesh	Panini Hans	11	71212	1
2	Andhra Pradesh	Girish Iyengar	9	69553	2
3	Andhra Pradesh	Sai Rajagopalan	9	69200	3
4	Andhra Pradesh	Manbir Rana	12	64035	4
5	Andhra Pradesh	Parth Biswas	12	58134	5
6	Arunachal Pradesh	Ijaya Dugal	9	75470	1
7	Arunachal Pradesh	Chanakya Dara	8	69985	2
8	Arunachal Pradesh	Maya Bhagat	12	67409	3
9	Arunachal Pradesh	Hemal Shroff	11	64678	4
10	Arunachal Pradesh	Nathaniel Sahni	9	63310	5
11	Assam	Indrajit Kalita	11	77116	1

Sales Trend Analysis

This section focuses on analyzing monthly sales trends to understand revenue growth patterns.

1. Monthly sales trends

	year numeric	month numeric	current_month_sales numeric	last_month_sales numeric
1	2025	4	8306086	[null]
2	2025	5	8610972	8306086
3	2025	6	8138007	8610972
4	2025	7	8527543	8138007
5	2025	8	8807553	8527543
6	2025	9	8340932	8807553
7	2025	10	8744039	8340932
8	2025	11	8209737	8744039
9	2025	12	9191809	8209737
10	2026	1	8426911	9191809
11	2026	2	7731046	8426911
12	2026	3	8204872	7731046

Return and Delivery Analysis

This section analyzes return behavior and delivery performance to identify operational issues

1. Most returned products

	product_name character varying (30)	status character varying (15)	return_rate numeric
1	Water Bottle	Returned	0.16
2	E-reader	Returned	0.15
3	Carrom Board	Returned	0.15
4	Hair Spray	Returned	0.14
5	Shaving Cream	Returned	0.14
6	Hair Oil	Returned	0.13
7	Eye Shadow	Returned	0.13
8	Utensil Set	Returned	0.13
9	Badminton Racket	Returned	0.13
10	Wall Clock	Returned	0.13

2. Most returned category

	category_name character varying (15)	delivery_status character varying (15)	return_rate numeric
1	Beauty	Returned	4.27
2	Clothing	Returned	4.25
3	Sports	Returned	4.24
4	Electronics	Returned	4.23
5	Home	Returned	4.19

Payment Analysis

This section evaluates payment success and failure rates.

1. Payment success rate

	payment_status character varying (20)	percentage numeric
1	Completed	69.93
2	Pending	20.05
3	Refunded	5.04
4	Failed	4.99

Seller Analysis

This section evaluates seller performance and identifies inactive sellers.

1. Top performing sellers

	seller_name character varying (40)	total_order bigint	total_sales numeric
1	Sarkar Group	335	1661291
2	Dada Group	190	1051738
3	Ahuja and Sons	189	1036956
4	Sen, Dhar and Kalla	187	1007344
5	Sehgal LLC	177	1006941

2. Inactive sellers

	seller_id [PK] character varying (10)	seller_name character varying (40)	last_order date

Inventory Analysis

This section identifies products with low stock levels to prevent stock shortages.

1. Inventory stock alert

	product_name character varying (30)	stock_left integer	warehouse character varying (10)	last_restock date
1	Cargo Pants	5	WH_22	2026-02-23
2	Knife Set	7	WH_25	2026-02-19
3	Boxing Gloves	4	WH_5	2025-10-02

Key Insights

- A Clothing category contributes a most in revenue
- A small group of customers generates the majority of total sales (high CLV impact)
- There are 3 customers with no orders
- Beauty categories have higher return rates, affecting profitability
- Sales trends show fluctuations, indicating seasonal demand patterns
- Seller performance is consistent, with no inactive sellers observed in the dataset